

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD  
CENTRAL VALLEY REGION

ORDER NO. R5-2002-0077

WASTE DISCHARGE REQUIREMENTS  
FOR  
WILD WINGS WATER RECYCLING FACILITY  
WILD WINGS, LLC  
YOLO COUNTY

The California Regional Water Quality Control Board, Central Valley Region, (hereafter Regional Board) finds that:

1. Wild Wings, LLC, submitted a Report of Waste Discharge (RWD) dated 29 June 2001 for the construction of a water recycling facility with reuse for golf course irrigation at the proposed Wild Wings development. The Wild Wings Water Recycling Facility (WWRF) will be owned and operated by the Wild Wings County Service Area or an appropriate maintenance/funding entity as approved by the Yolo County Board of Supervisors. The golf course will be owned and operated by the Wild Wings, LLC. The RWD contained a recycled water management plan, but contained insufficient information on shallow groundwater quality at the proposed facility. By letter dated 25 July 2001, the RWD was found to be incomplete. A workplan for a groundwater investigation was subsequently submitted and on 28 August 2001 the RWD was considered complete. The report of the shallow groundwater sampling was submitted on 5 October 2001.
2. The proposed Wild Wings Development will be located on land currently owned by Milt Watts, et al. Concurrently with the recordation of the first final subdivision map for the Wild Wings Development, said land will transfer from Milt Watts, et al. to the Wild Wings, LLC. The Wild Wings, LLC, shall cause the design, construction and permitting of the WWRF. Construction of the WWRF will not begin prior to the transfer of the land from Milt Watts, et al, to Wild Wings, LLC. Prior to the commencement of operations of the WWRF, the ownership and control of the WWRF shall be transferred to the Wild Wings County Service Area, or the appropriate maintenance/funding entity as approved by the Yolo County Board of Supervisors.
3. Wild Wings, LLC is hereafter referred to as Discharger.
4. On 29 June 2001, the Discharger submitted an engineering report for the use of recycled water entitled *Engineering Report for Production, Distribution, and Use of Recycled Water* (Engineering Report). The Engineering Report was transmitted to the California Department of Health Services (DHS) on 24 July 2001 for a review of compliance with Section 60323, Title 22, California Code of Regulations (CCR). The Regional Board has consulted with DHS and has implemented or addressed their comments and recommendations received by phone in January 2002 and by letter dated 8 March 2002.

5. The Wild Wings development will consist of 337 homes, a nine-hole golf course, and a WWRF. The community will be approximately five miles west of the City of Woodland, adjacent to the Woodland-Watts Airport, near the intersection of Highway 16 and County Road 94B in an unincorporated area of Yolo County. Cache Creek transects the northwestern portion of the property. The development will encompass approximately 238 acres (Assessor's Parcel Nos. 025-440-017 and 025-440-043), north of Section 32 in T10N, R1E, MDB&M. The location of the development, WWRF and recycled water application areas are shown on Attachments A, B, C, D, E and F which are attached hereto and are part of this Order by reference.
6. Full build-out of the development is anticipated to occur in three phases over a period of about five years. The initial phase will consist of 77 units and the second and third phases will each consist of 130 units. The golf course is proposed for the second phase of construction. Until the golf course is constructed, the area will remain as agricultural land and will be irrigated with recycled water.
7. The WWRF will occupy 3.67 acres (400 by 400 feet) along the western side of the development (see Attachment C). Domestic wastewater generated from the development will be treated to a tertiary level at the WWRF for recycling. The recycled water will be blended with groundwater obtained from an on-site supply well, then used to irrigate the Wild Wings golf course and adjacent landscaped area (also jointly referred to as recycled water application area), which occupies 93 acres, as shown on Attachment F. When water supplies are adequate, surface water from Moore Canal, which crosses the northwestern portion of the property, may be used for blending with the recycled water for irrigation purposes. Recycled water or recycled water blended with groundwater or surface water will not be allowed to discharge or runoff from the application areas to surface waters, surface water drainage courses or to water features within the golf course.
8. Wastewater flows are estimated to be 101,000 gallons per day (gpd), or about 70 gallons per minute (gpm), based on a daily wastewater generation rate of 300 gallons per household per day. Peak hourly wastewater flows are estimated to be 246 gpm, based on an estimated peaking factor of 3.5.
9. The WWRF will consist of an influent pump station, an extended aeration biological package treatment plant, a tertiary filtration system, a chlorine disinfection unit, a recycled water storage pond, and an irrigation pump station. Wastewater will be treated to meet Title 22, CCR, Section 60301 et. seq. (hereafter Title 22) standards for unrestricted use. The recycled water storage pond will be lined with a reinforced polypropylene, ultra-violet treated, 45-mil liner and will be used for the storage of the recycled water. The entire WWRF will be landscaped and fenced to reduce visibility and access. The golf maintenance facility will be located within the limits of the WWRF. The site plan and process schematic are shown on Attachments D and E, respectively.

10. The WWRF will include features to ensure design reliability, including additional aeration and sedimentation basin capacity, a stand-by emergency power generator, and an emergency storage tank for diversion of wastewater effluent that does not meet the required treatment levels. The emergency storage tank(s) will have an approximate storage capacity of 100,000 gallons, which is equal to approximately one days wastewater flow, at complete build-out. If water is diverted to the emergency storage tank, it will be returned to the influent pump station for re-treatment. The WWRF will also be equipped with an alarm system with autodialer, pager, and uninterrupted power supply that will be activated during irregular operating conditions.
11. The WWRF will have a built-in aerated equalization tank at the beginning of the bioprocess equipment that includes a screen. The screen will be manually raked each day and collected screenings will be stored onsite in a container and hauled offsite weekly and taken to a permitted facility for disposal.
12. The WWRF will also include a built-in sludge concentrator that will serve as a digester and a holding tank. Waste digested sludge from this process will be transported offsite once per week and taken to a permitted facility for disposal and/or treatment.
13. Raw and treated wastewater are estimated to have the following characteristics:

| <u>Compound</u>                                                | <u>Units</u> | <u>Wastewater Concentration</u> |                |
|----------------------------------------------------------------|--------------|---------------------------------|----------------|
|                                                                |              | <u>Raw</u>                      | <u>Treated</u> |
| Biochemical Oxygen Demand <sub>5-Day</sub> (BOD <sub>5</sub> ) | mg/l         | 300                             | <10            |
| Total Suspended Solids (TSS)                                   | mg/l         | 300                             | <10            |
| Settleable Solids                                              | ml/l         | 10                              | <1             |
| Total Nitrogen                                                 | mg/l         | 40                              | 30             |
| Total Dissolved Solids (TDS) <sup>1</sup>                      | mg/l         | 765                             | 765            |
| Total Coliform Organisms (Total Coliform)                      | MPN/100ml    | 10 <sup>8</sup>                 | <2.2           |

<sup>1</sup> The concentration is not expected to reduce as a result of treatment in the extended aeration plant.

mg/l = Milligrams per liter

ml/l = Milliliters per liter

MPN/100 ml = Most Probable Number per 100 Milliliters

< or ≤ = less than or less than or equal to the listed reporting limit

14. Groundwater from on-site supply wells and/or surface water (if available from Moore Canal) will be used to supplement the recycled water for irrigating the recycled water application area. The supplemental water will be discharged through an air gap to the lined storage pond. Once the recycled water is blended with groundwater or surface water, the blended water shall be identified as recycled water. The mix ratio of recycled water and supplemental ground water or surface water will vary by month, but is estimated to be about 27 percent treated wastewater

during normal precipitation years, supplying approximately one-fourth of the golf course irrigation need.

15. The storage pond will be designed to provide a 2-foot freeboard during 100-year precipitation conditions. The pond will be approximately 250 by 350 feet (pond surface area of 1.7 acres) with a maximum depth of 20 feet near the center of the pond. The holding capacity of the pond is approximately 7.5 million gallons. The pond will be excavated 20 feet below grade and will be surrounded by a 2-foot berm with interior and exterior slopes of 2:1.
16. Blended recycled water in the storage pond will be pumped during off hours (generally from 10 pm to 4 am) and with irrigation applications occasionally occurring from 8 pm to 6 am. Irrigation will not occur during precipitation events or during periods of excessive wind. Low trajectory sprinkler heads will be used to reduce aerosol drift. The blended, recycled water will be applied up to the property line of the residential parcels, which will be contoured to drain toward the golf course. Warning signs regarding the use of recycled water will be posted.
17. The Wild Wings development will include setbacks for the WWRF and for the recycled water areas.
18. The RWD provided two water balances for the WWRF for normal-year and 100-year precipitation conditions. The storage pond is estimated to have sufficient capacity for the treated/blended wastewater during 100-year precipitation conditions. During normal year conditions, a total of 410 acre-feet (ac-ft) of water will be applied to the golf course with an estimated percolate depth is 20.5 inches; whereas, only 396 ac-ft of water will be applied for 100-year precipitation conditions with an estimated percolate depth is 31.7 inches. During these latter conditions, the maximum monthly percolation rate is estimated to be 0.0087 inches per hour (in/hr), where the limiting percolation rate for the on-site soil is 0.06 in/hr. The golf course is expected to have sufficient capacity for recycled water applications, even during 100-year precipitation conditions.
19. The annual BOD loading rate is estimated to be 32.4 pounds per acre-year (lb/ac-yr), with a maximum loading rate 0.12 pounds per acre per day occurring in the summer when application rates are the greatest. The annual nitrogen load is estimated to be 97 lb/ac-yr, which is less than the nitrogen uptake rate for turf (over 200 lb/ac-yr). Therefore, fertilizers may be needed to meet turf nutrient demands for the golf course.
20. All storm water runoff generated at the WWRF will be contained onsite. All storm water runoff generated within the Wild Wings development will pass through the golf course detention system, consisting of swales and connected lakes that will be linked by pipelines. Mosquito fish will be stocked in the golf course lakes, which will be used for storage and surge protection. Outflow from the lakes will occur by gravity when the lake level reaches a

designated elevation with discharge via pipeline to Cache Creek and surface channel to Moore Canal.

21. The golf course will be required to manage the application of recycled water such that recycled water does not co-mingle with stormwater and be discharged to surface water or surface water drainage courses. The golf course may use course design, engineering controls and/or operational controls to prevent the discharge of recycled water or co-mingled recycled water and stormwater to surface water or surface water drainage courses. If recycled water and stormwater are allowed to co-mingle they must be contained on-site or be discharged under a site-specific National Pollution Discharge Elimination System (NPDES) permit approved by the Regional Board.
22. Characterization of background water quality for Moore Canal was not provided in the RWD. However, the Monitoring and Reporting Program requires surface water monitoring upstream and downstream of the golf course and development.
23. Precipitation and evapotranspiration data, collected at the Davis California Irrigation Management Information System (CIMIS) station since 1983, was provided in the RWD. The average precipitation is 19.3 inches and the average evapotranspiration is 56.9 inches.
24. The Wild Wings development is divided into two main watersheds, located east and west. Drainage from the western portion of the development drains northerly to Cache Creek through an existing channel located between Wild Wings and the adjacent Yolo Fliers Club. The drainage is routed through a pipeline under Moore Canal. The eastern portion of the development drains northeasterly to Moore Canal. Easterly surface water drainage will be through an existing channel located north of the Yolo Fliers Club airport and adjacent to County Road 94B.
25. Water supply for the development and the wastewater facility will be obtained from two supply wells as shown on Attachment F. Characterization of the deep (300 to 700 feet below ground surface) groundwater quality was obtained from a supply well sampled in April 2001 at the Yolo Fliers Club and is summarized as follows:

| <u>Compound</u>               | <u>Units</u> | <u>Concentration</u> |
|-------------------------------|--------------|----------------------|
| pH                            | pH Units     | 8.2                  |
| Electrical Conductivity (EC)  | µmhos/cm     | 790                  |
| TDS                           | mg/l         | 490                  |
| Chloride                      | mg/l         | 45                   |
| Nitrate (as NO <sub>3</sub> ) | mg/l         | <2.0                 |
| Sulfate                       | mg/l         | 83                   |
| Copper                        | µg/l         | <50                  |
| Iron                          | µg/l         | <100                 |

| <u>Compound</u> | <u>Units</u> | <u>Concentration</u> |
|-----------------|--------------|----------------------|
| Manganese       | µg/l         | 78                   |
| Zinc            | µg/l         | <50                  |

µmhos/cm = Micromhos per centimeter

NO<sub>3</sub> = Nitrate

< = Concentration was not detected above the listed reporting limit.

26. Background shallow groundwater was characterized at the proposed site of the WWRF based on samples collected from four borehole locations (HB-2, HB-3, HB-7 and B-27), as shown on Attachment F. Shallow groundwater samples were collected at depths from 33 to 38 feet bgs from these borings. The results of these background groundwater samples are provided below.

| <u>Compound</u>        | <u>Units</u> | <u>Boring</u> |             |             |             | <u>Average</u> |
|------------------------|--------------|---------------|-------------|-------------|-------------|----------------|
|                        |              | <u>HB-2</u>   | <u>HB-3</u> | <u>HB-7</u> | <u>B-27</u> |                |
| pH                     | pH Units     | 7.5           | 7.6         | 7.8         | 7.6         | 7.6            |
| EC                     | µmhos/cm     | 1400          | 1400        | 1500        | 1500        | 1450           |
| TDS                    | mg/l         | 1100          | 1100        | 1100        | 1100        | 1100           |
| Fixed Dissolved Solids | mg/l         | 840           | 865         | 850         | 880         | 859            |
| Ammonia (as N)         | mg/l         | 0.34          | 0.30        | 0.82        | 0.55        | 0.50           |
| Chloride               | mg/l         | 130           | 130         | 150         | 140         | 138            |
| Nitrate (as N)         | mg/l         | 16            | 18          | 19          | 19          | 18             |
| Sulfate                | mg/l         | 190           | 190         | 190         | 180         | 188            |
| Copper                 | mg/l         | <0.05         | <0.05       | <0.05       | <0.05       | <0.05          |
| Iron                   | mg/l         | <0.1          | <0.1        | <0.1        | <0.1        | <0.1           |
| Manganese              | mg/l         | 1.1           | <0.03       | 0.16        | 1.3         | 0.64           |
| Sodium                 | mg/l         | 150           | 150         | 160         | 160         | 155            |
| Zinc                   | mg/l         | <0.05         | <0.05       | <0.05       | <0.05       | <0.05          |
| Total Coliform         | MPN/100ml    | <2.0          | <2.0        | <2.0        | <2.0        | <2.0           |
| Fecal Coliform         | MPN/100ml    | <2.0          | <2.0        | <2.0        | <2.0        | <2.0           |

(as N) = as Nitrogen

27. Depth to shallow groundwater at the development ranges between about 35 to 75 feet bgs. The RWD estimated the groundwater flow direction to be southerly.
28. Twenty-four soil borings were installed to a depth of about 16 feet bgs to assess soil conditions at the Wild Wings project. Five different soil types are found across the project, with the San Ysidro loam and the Hillgate loam predominating. All of the soil types have permeabilities ranging from moderate to very slow and with drainage described as moderate to good. Most of the soil types have a silty-clay or clay subsoil with an underlying coarser soil (ranging from a sandy silt to a gravelly sand) that is generally encountered between 5 to 10 feet bgs.

29. Local land use consists primarily of interspersed agriculture, open space, and low-density residential to the west and south of the proposed development. The Woodland-Watts Airport borders the development on the east and the Yolo Fliers Club and golf course are to the north of the proposed development. The Moore Canal and Cache Creek lie further to the north of the project (generally less than ½ mile) and cross the northwestern portion of the development property. The WWRF for the development is approximately 2800 and 3500 feet south of the canal and creek, respectively.
30. The ground surface elevation at the Wild Wings development generally ranges between 120 and 125 feet mean sea level (msl). The WWRF will be installed at an elevation of about 121 feet msl.

### **Sanitary Sewer System Description**

31. The Discharger's sanitary sewer system collects wastewater using sewers, pipes, pumps, and/or other conveyance systems and directs this raw sewage to the WWRF. A "sanitary sewer overflow" is defined as a discharge to ground or surface water from the sanitary sewer system at any point upstream of the WWRF. Temporary storage and conveyance facilities, such as wet wells, regulated impoundments, tanks and highlines are part of the sanitary sewer system and discharges to these facilities are not considered sanitary sewer overflows, provided that the waste is fully contained within these temporary storage/conveyance facilities.
32. Sanitary sewer overflows consist of varying mixtures of domestic sewage, industrial wastewater, and commercial wastewater; this mixture depends on the pattern of land use in the sewage collection system tributary to the overflow. The primary causes of sanitary sewer overflows include grease blockages, root blockages, debris blockages, sewer line flood damage, manhole structure failures, vandalism, pump station mechanical failures, power outages, storm or groundwater inflow/infiltration, lack of capacity, and contractor-caused blockages.
33. Sanitary sewer overflows often contain high levels of suspended solids, pathogenic organisms, toxic pollutants, nutrients, oxygen demanding organic compounds, oil and grease, and other pollutants. Sanitary sewer overflows can cause temporary exceedences of applicable water quality objectives, pose a threat to public health, adversely affect aquatic life, and impair the public recreational use and aesthetic enjoyment of surface waters in the area.
34. The Discharger is expected to take all necessary steps to adequately maintain, operate, and prevent unintended discharges from its sanitary sewer collection system. This Order requires the Discharger to prepare and implement best practicable treatment and control (BPTC) to prevent unintended discharges from the sanitary sewer collection system. The Regional Board acknowledges that this is a new system; however, Regional Board experience has shown that a *Sanitary Sewer Collection Operation, Maintenance, Overflow Prevention, and Response Plan (SSO Plan)* and good Operation and Maintenance (O&M) may prevent future overflows.

Therefore, the Discharger is required to develop an O&M Plan and SSO Plan pursuant to a compliance schedule as outlined in Provisions G.5 and G.12.

### **Treatment and Control Practices**

35. The WWRF provides BPTC of the discharge that incorporates:
- a. technology for secondary treatment of municipal wastewater;
  - b. technology for tertiary treatment of municipal wastewater;
  - c. disinfection;
  - d. recycling of wastewater on a golf course and landscape areas; and
  - e. lining of the wastewater storage pond.
36. Depth to groundwater is estimated to be approximately 35 feet bgs at the WWRF, golf course and landscape areas, thereby providing a soil media that can provide additional treatment and/or attenuation of constituents of concern. However, the Regional Board has incorporated groundwater monitoring at the WWRF to ensure that the BPTC, as outlined in Finding No. 35 of this Order, incorporated by the Discharger will provide adequate protection of groundwater. If it is determined that the discharge is causing groundwater quality to be degraded beyond background water quality and exceeds the Groundwater Limitations set forth in this Order, then the Discharger, upon direction of the Executive Officer, will be required to complete the tasks as required in the Provisions G.14a through G.14.d of this Order, and implement the approved strategies developed from that work. This implementation will ensure that BPTC and the highest water quality consistent with the maximum benefit to the people of the State will be achieved.

### **Water Recycling**

37. The DHS requires that the American Water Works Association (AWWA) Guidelines for Distribution of Non-Potable Water be implemented in design and construction of recycling equipment. The guidelines require installation of purple pipe, adequate signs, etc. Adequate separation of the recycled water lines, the domestic water lines, and the sewer lines shall be provided.
38. DHS has established statewide water recycling criteria in Title 22. DHS revised the water recycling criteria contained in Title 22 on 2 December 2000. The revised Title 22 required that all wastewater used for reclamation receive, at a minimum, secondary treatment. However, Title 22 does not define secondary treatment with respect to numerical limits for BOD<sub>5</sub> and TSS. According to DHS, for uses not requiring disinfection, treated wastewater should, at a minimum, be adequately oxidized, contain dissolved oxygen near saturation levels, and be nonputrescible when applied to land.



39. DHS requirements for recycled water for use as golf course irrigation generally require a higher level of treatment than use of recycled water for irrigation of pasture. Golf course irrigation generally requires that the recycled water be treated to tertiary standards with disinfection to 2.2 MPN/100 ml. Irrigation of pasture can be conducted with recycled water disinfected to 23 MPN/100 ml.
40. Section 60303 of Title 22 states that water recycling requirements shall not apply to the use of recycled water on-site at a water recycling plant, or wastewater treatment plant, provided access by the public to the area of on-site recycled water use is restricted.
41. Section 60323(a) Title 22 states that no person shall produce or supply recycled water for direct reuse from a water recycling plant unless an engineering report is submitted for review and approval by DHS and the Regional Board. On 29 June 2001, the Regional Board received the Title 22 report. On 24 July 2001, the report was forwarded to DHS for review. Verbal and written comments were received by DHS in January 2002 and March 2002, respectively, and incorporated into this Order

### **Groundwater Degradation**

42. State Water Resources Control Board (State Board) Resolution No. 68-16 (hereafter Resolution 68-16 or the "Antidegradation Policy") requires the Regional Board in regulating the discharge of waste to maintain high quality waters of the state until it is demonstrated that any change in quality will be consistent with maximum benefit to the people of the State, will not unreasonably affect beneficial uses, and will not result in water quality less than that described in the Regional Board's policies (i.e., quality that exceeds water quality objectives).
43. The Regional Board finds that some degradation of groundwater beneath the WWRF (excluding the recycling water application areas) is consistent with Resolution 68-16 provided that:
  - a. The degradation is confined to a specified area;
  - b. The discharger minimizes the degradation by fully implementing, regularly maintaining, and optimally operating BPTC measures;
  - c. The degradation is limited to waste constituents typically encountered in municipal wastewater as specified in the groundwater limitations in this Order; and
  - d. The degradation does not result in water quality less than that prescribed in the Basin Plan.
44. Some degradation of groundwater by some of the typical waste constituents released with discharge from a municipal wastewater facility after effective source control, treatment and control is consistent with maximum benefit to the people of California. The technology, energy, water recycling, and waste management advantages of municipal utility service far exceed any benefits derived from a community otherwise reliant on numerous concentrated

individual wastewater systems, and the impact on water quality will be substantially less. Degradation of groundwater by constituents (e.g., toxic chemicals) other than those specified in the groundwater limitations in this Order, and by constituents that can be effectively removed by conventional treatment (e.g., total coliform bacteria), is prohibited. When allowed, the degree of degradation permitted depends upon many factors (i.e., background water quality, the waste constituent, the beneficial uses and most stringent water quality objective, source control measures, waste constituent treatability).

45. Economic prosperity of local communities and associated industry is of maximum benefit to the people of California, and therefore sufficient reason exists to accommodate growth and groundwater degradation around the wastewater treatment plant, provided that the terms of the Basin Plan are met.
46. These waste discharge requirements (WDRs) do not allow degradation of groundwater beneath the recycled water application areas. The Discharger is not required to monitor the groundwater under the recycled water areas due to the advanced tertiary treatment of the wastewater.
47. This Order establishes interim groundwater limitations that will not unreasonably threaten present and anticipated beneficial uses or result in additional groundwater degradation. This Order contains tasks for assuring that BPTC and the highest water quality consistent with the maximum benefit to the people of the State will be achieved. Accordingly, the discharge is consistent with the antidegradation provisions of Resolution 68-16. Based on the results of the scheduled tasks, the Regional Board may reopen this Order to reconsider groundwater limitations and other requirements to comply with Resolution 68-16.

#### **Beneficial Uses**

48. Surface water drainage in the area of the development is generally to the north toward Cache Creek, and to the northeast toward Moore Canal. Cache Creek drains to the Yolo Bypass. Moore Canal drains to Willow Slough, which then drains to the Yolo Bypass. The site is in the Lower Cache Creek Hydrologic Area (No. 511.30), as depicted on interagency hydrologic maps prepared by the Department of Water Resources in August 1986.
49. The beneficial uses of the Moore Canal, Cache Creek and the Yolo Bypass are agricultural supply for irrigation and stock watering, contact and non-contact recreation, warm freshwater habitat, fish migration and spawning, wildlife habitat and groundwater recharge.
50. The beneficial uses of the underlying groundwater are municipal and domestic supply, agricultural supply, industrial service supply and industrial process supply.

### **Basin Plan and Regulatory Considerations**

51. The monitoring and reporting program required by this Order and the attached “Monitoring and Reporting Program No. R5-2002-0077” are necessary to assure compliance with these waste discharge requirements. The Discharger operates the facility that discharges the waste subject to this Order.
52. The State Board adopted Resolution No. 77-1, Policy with Respect to Water Reclamation in California, which finds that the State shall undertake all possible steps to encourage the development of water reclamation facilities so that recycled water may be made available to help meet the growing water requirements of the State. This policy provides that the Regional Board shall encourage water reclamation projects which do not adversely impact vested water rights, or unreasonably impair in-stream beneficial uses, or place an unreasonable burden on present water supply systems in cases where recycled water will replace or supplement the use of fresh water or better water quality.
53. The *Water Quality Control Plan for the Sacramento River and San Joaquin River Basins, Fourth Edition*, (hereafter Basin Plan) designates beneficial uses, establishes water quality objectives, contains implementation plans and policies for protecting waters of the basin, and incorporates by reference plans and policies adopted by the State Board. Pursuant to Section 13263(a) of the California Water Code, waste discharge requirements must implement the Basin Plan.
54. The Basin Plan encourages water recycling.
55. The Basin Plan establishes numerical and narrative water quality objectives for surface and groundwater within the basin, and recognizes that water quality objectives are achieved primarily through the Regional Board’s adoption of waste discharge requirements and enforcement orders. Where numerical water quality objectives are listed, these are limits necessary for the reasonable protection of beneficial uses of the water. Where compliance with narrative water quality objectives is required, the Regional Board will, on a case-by-case basis, adopt numerical limitations in orders, which will implement the narrative objectives to protect beneficial uses of the waters of the state.
56. The Basin Plan identifies numerical water quality objectives for waters designated as municipal supply. These are the maximum contaminant levels (MCLs) specified in the following provisions of Title 22, CCR: Tables 64431-A (Inorganic Chemicals) and 64431-B (Fluoride) of Section 64431, Table 64444-A (Organic Chemicals) of Section 64444, and Table 64449-A (Secondary Maximum Contaminant Levels-Consumer Acceptance Limits) of Section 64449. The Basin Plan’s incorporation of these provisions by reference is prospective, and includes future changes to the incorporated provisions as the changes take effect. The Basin Plan recognizes that the Regional Board may apply limits more stringent than MCLs to ensure that

waters do not contain chemical constituents in concentrations that adversely affect beneficial uses.

57. The Basin Plan contains narrative water quality objectives for chemical constituents, tastes and odors, and toxicity. The chemical constituent objective requires that groundwater shall not contain chemical constituents in concentrations that adversely affect beneficial uses. The taste and odor objective requires that groundwater shall not contain tastes and odors producing substances in concentrations that cause a nuisance or adversely affects beneficial uses. The toxicity objective requires that groundwater be maintained free of toxic substances in concentrations that produce detrimental physiological responses in humans, plants or animals.
58. The Basin Plan indicates that achievement of water quality objectives depends on the application of the objectives to controllable water quality factors. Controllable water quality factors consist of actions, conditions, or circumstances resulting from human activities that may influence the quality of the waters of the State that may be reasonably controlled. Controllable factors are not allowed to cause further degradation of water quality in instances where uncontrollable factors have already resulted in water quality objectives being exceeded.
59. Water quality criteria that are protective of surface and/or groundwater beneficial uses (i.e., water quality objectives) are as follows:

| <u>Compound</u>                            | <u>Units</u> | <u>Concentrations</u> <sup>1</sup>  |
|--------------------------------------------|--------------|-------------------------------------|
| pH <sup>2</sup>                            | pH Units     | 6.5 to 8.5                          |
| EC <sup>3</sup>                            | µmhos/cm     | 700 <sup>4</sup> , 900, 1600, 2200  |
| TDS <sup>3</sup>                           | mg/l         | 450 <sup>4</sup> , 500, 1000, 1500  |
| Total coliform                             | MPN/100 ml   | 2.2 <sup>1</sup>                    |
| Chloride                                   | mg/l         | 106 <sup>4</sup> , 250 <sup>2</sup> |
| Nitrate (as Nitrogen/as Nitrate)           | mg/l         | 10 <sup>5</sup> /45 <sup>5</sup>    |
| Sulfate                                    | mg/l         | 250 <sup>2</sup> , 500 <sup>5</sup> |
| Boron                                      | mg/l         | 0.7 <sup>4</sup>                    |
| Copper                                     | mg/l         | 1.0 <sup>2</sup> , 1.3 <sup>5</sup> |
| Iron                                       | mg/l         | 0.3 <sup>2</sup>                    |
| Manganese                                  | mg/l         | 0.05 <sup>2</sup>                   |
| Sodium                                     | mg/l         | 2.0 <sup>6</sup>                    |
| Zinc                                       | mg/l         | 5.0 <sup>2</sup>                    |
| Ammonia                                    | mg/l         | 0.5 <sup>7</sup>                    |
| Total Trihalomethanes (TTHMs) <sup>8</sup> | mg/l         | 0.1 <sup>9</sup> /0.08 <sup>9</sup> |

<sup>1</sup> Source = *A Compilation of Water Quality Goals*, California Environmental Protection Agency, Regional Water Quality Control Board, Central Valley Region, August 2000.

<sup>2</sup> Secondary Maximum Contaminant Level

- <sup>3</sup> Recommended level, Upper-level, and Short-term level, respectively for Secondary Maximum Contaminant Level (MCL).
- <sup>4</sup> Agricultural Water Quality Goal
- <sup>5</sup> Primary MCL.
- <sup>6</sup> Suggested No-Adverse Response Level (SNARL)
- <sup>7</sup> *On the Quality of Water Intended for Human Consumption*, Council of the European Union, Council Directive 98/83/EC, 3 November 1998.
- <sup>8</sup> TTHMs consist of the sum of bromodichloromethane, bromoform, chloroform, and dibromochloromethane.
- <sup>9</sup> Effective 12/17/01 Primary MCL is 0.1 mg/l for systems serving greater than 10,000 people. The Primary MCL will be 0.08mg/l for all other systems on 12/17/03.

60. Section 13241 of the California Water Code (CWC) requires the Regional Board to consider various factors, including economic considerations, when adopting water quality objectives into its Basin Plan. Water Code Section 13263 requires the Regional Board to address the factors in Section 13241 in adopting WDRs. The State Board, however, has held that a Regional Board need not specifically address the Section 13241 factors when implementing existing water quality objectives in WDRs because the factors were already considered in adopting water quality objectives. These WDRs implement adopted water quality objectives. Therefore, no additional analysis of Section 13241 factors is required.
61. Section 13267(b)(1) of the CWC provides that: *“In conducting an investigation specified in subdivision (a), the regional board may require that any person who has discharged, discharges, or is suspected of having discharged or discharging, or who proposes to discharge waste within its region, or any citizen or domiciliary, or political agency or entity of this state who has discharged, discharges, or is suspected of having discharged or discharging, or who proposes to discharge, waste outside of its region that could affect the quality of waters within its region shall furnish, under penalty of perjury, technical or monitoring program reports which the regional board requires. The burden, including costs, of these reports shall bear a reasonable relationship to the need for the report and the benefits to be obtained from the reports. In requiring those reports, the regional board shall provide the person with a written explanation with regard to the need for the reports, and shall identify the evidence that supports requiring that person to provide the reports.”*

The technical reports required by this Order and the attached “Monitoring and Reporting Program No. R5-2002-0077” are necessary to assure compliance with these waste discharge requirements. The Discharger operates the facility that discharges the waste subject to this Order.

62. The United States Environmental Protection Agency (US EPA) has promulgated biosolids reuse regulations in 40 Code of Federal Regulations (CFR) Part 503, *Standard for the Use or Disposal of Sewage Sludge*, which establishes management criteria for protection of ground and surface waters, sets application rates for heavy metals, and establishes stabilization and disinfection criteria.

63. The Regional Board is using the Standards in 40 CFR 503 as guidelines in establishing this Order, but the Regional Board is not the implementing agency for 40 CFR 503 regulations. The Discharger may have separate and/or additional compliance, reporting, and permitting responsibilities to the US EPA.
64. The State Board adopted Order No. 97-03-DWQ (General Permit No. CAS000001) specifying waste discharge requirements for discharges of storm water associated with industrial activities. This Order requires the submittal of a NOI or Notice of Non-Applicability by all affected industrial dischargers.
65. California Department of Water Resources standards for the construction and destruction of groundwater wells (hereafter DWR Well Standards), as described in *California Well Standards Bulletin 74-90* (June 1991) and *Water Well Standards: State of California Bulletin 94-81* (December 1981), and any more stringent standards adopted by the Discharger or county pursuant to CWC section 13801, apply to all monitoring wells.
66. State regulations that prescribe procedures for detecting and characterizing the impact of waste constituents from waste management units on groundwater are found in Title 27. While the WWRF is exempt from Title 27, the data analysis methods of Title 27 are appropriate for determining whether the discharge complies with the terms for protection of groundwater specified in this Order.
67. The discharge authorized herein and the treatment and storage facilities associated with the discharge, except for discharges of residual sludge and solid waste, are exempt from the requirements of Title 27, California Code of Regulations (CCR), Section 20005 et seq. (hereafter Title 27). The exemption, pursuant to Section 20090(a) of Title 27, is based on the following:
  - a. The waste consists primarily of domestic sewage and treated effluent;
  - b. The waste discharge requirements are consistent with water quality objectives; and
  - c. The treatment and storage facilities described herein are associated with a municipal wastewater treatment plant.
68. Pursuant to CWC Section 13263(g), discharge is a privilege, not a right, and adoption of this Order does not create a vested right to continue the discharge.

### **Public Notice**

69. The Regional Board considered all the above and the supplemental information and details in the attached Information Sheet, which is incorporated by reference herein, in establishing the following conditions of discharge.

70. The Regional Board consulted with the DHS and has considered and implemented their verbal and written comments and recommendations regarding the public health aspects of water recycling.
71. On 5 May 1992, the Yolo County Board of Supervisors issued a Notice of Exemption, which determined that the Wild Wings development is consistent with the adopted Specific Plan for Monument Hill Special Study. An Environmental Impact Report (EIR) for the Specific Plan for the Monument Hill Special Study was certified on 20 August 1985. The Regional Board reviewed the EIR and concurs that reuse of highly treated recycled water, pond lining, and other BPTC measures will mitigate the risk of adverse impacts to groundwater.
72. The action to update waste discharge requirements for this facility is exempt from the provision of the California Environmental Quality Act (CEQA), in accordance with Section 15301, Title 14, CCR.
73. In the event of any change in control or ownership of land or waste discharge facilities presently owned or controlled by the Discharger, the Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be immediately forwarded to this office.  
To assume operation under this Order, the succeeding owner or operator must apply in writing to the Executive Officer requesting transfer of the Order. The request must contain the requesting entity's full legal name, the state of incorporation (if a corporation), the address and telephone number of the persons responsible for contact with the Regional Board and a statement. The statement shall comply with the signatory requirements of the Regional Board and state that the new owner or operator assumes full responsibility for compliance with this Order. Failure to submit the request shall be considered a discharge without requirements, a violation of the CWC. Transfer shall be approved or disapproved in writing by the Executive Officer.
74. The Regional Board has notified the Discharger and interested agencies and persons of its intent to prescribe waste discharge requirements for this discharge, and has provided them with an opportunity for a public hearing and an opportunity to submit their written views and recommendations.
75. The Regional Board, in a public meeting, heard and considered all comments pertaining to the discharge.

**IT IS HEREBY ORDERED** that, pursuant to Sections 13263 and 13267 of the California Water Code, that the Wild Wings, LLC, its agents, successors, and assigns, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted hereunder, shall comply with the following:

*[Note: Other prohibitions, conditions, definition and some methods of determining compliance are contained in the attached "Standard Provisions and Reporting Requirements for Waste Discharge Requirements," dated 1 March 1991, which are part of this Order. This attachment and its individual paragraphs are referred to as "Standard Provisions" and are attached hereto and made part of this Order by reference.]*

**A. Discharge Prohibitions:**

1. Direct or indirect discharge of wastes and/or recycled water to surface waters, surface water drainage courses or to golf course water features is prohibited.
2. Discharge of wastewater to the WWRF is prohibited until the Wild Wings County Service Area or an appropriate maintenance/funding entity as approved by the Yolo County Board of Supervisors obtains administration and control of the WWRF.
3. Application of recycled water to areas other than the designated recycled water areas is prohibited.
4. Bypass or overflow of untreated or partially treated waste is prohibited, except as discussed in the Standard Provisions.
5. Recycled water that is applied at rates in excess of the vegetation nitrogen requirements, or at rates that would cause excess nitrogen, dissolved solids, or metals to leach to groundwater is prohibited.
6. Discharge of waste classified as 'hazardous,' defined in Section 20164 of Title 27, CCR, or 'designated', as defined in Section 13173 of the CWC, is prohibited.
7. Discharge of any wastewater other than that from domestic sources or domestic equivalent is prohibited.
8. Discharge of septage to the sanitary sewer system or the WWRF is prohibited.
9. Surfacing of wastewater outside or downgradient of the storage pond is prohibited.

**B. Discharge Specifications:**

1. The 30-day average daily dry weather flow shall not exceed 101,000 gallons.
2. All systems and equipment shall be operated to maximize treatment of wastewater and optimize the quality of the discharge, including the treatment of wastewater such that it complies with Title 22 CCR, Section 60301.230 ("Disinfected Tertiary Recycled Water").



3. Only tertiary treated effluent and/or tertiary effluent combined with supplemental irrigation water shall be discharged to the recycled water storage pond.
4. Recycled water discharged from the treatment facility to the golf course or approved landscape areas shall not exceed the following limits:

| <u>Constituent</u>       | <u>Units</u> | <u>Daily<br/>Maximum</u> | <u>30-Day<br/>Average</u> | <u>30-Day<br/>Median</u> | <u>Daily<br/>Maximum</u> |
|--------------------------|--------------|--------------------------|---------------------------|--------------------------|--------------------------|
| BOD                      | mg/l         | ---                      | <10                       | ---                      | ---                      |
| TSS                      | ml/l         | ---                      | <10                       | ---                      | ---                      |
| Total Coliform Organisms | MPN/100ml    | ---                      | ---                       | 2.2                      | 23                       |
| Turbidity                | NTU          | 2                        | ---                       | ---                      | 10                       |
| TDS                      | mg/l         | ---                      | 765                       | ---                      | ---                      |
| Total Nitrogen           | mg/l         | ---                      | 30                        | ---                      | ---                      |

NTU = Nephelometric Turbidity Unit

5. An alarm system shall be installed to provide notification when the treated effluent does not meet treatment standards.
6. All effluent that does not meet the secondary and tertiary treatment standards shall be routed to the emergency storage tank.
7. The treatment and disposal facilities shall be designed, constructed, operated, and maintained to prevent inundation or washout due to floods with a 100-year return frequency.
8. The wastewater ponds shall have sufficient capacity to accommodate allowable wastewater flow, design seasonal precipitation, and ancillary inflow and infiltration. Design seasonal precipitation shall be based on total annual precipitation using a return period of 100 years, distributed monthly in accordance with the historical rainfall patterns.
9. The freeboard in the recycled water storage ponds shall never be less than two feet, measured vertically to the lowest point of the berm.
10. On or about **1 October** each year, available pond storage capacity shall at least equal the volume necessary to comply with Discharge Specifications No. 8 and No. 9.
11. The following setbacks for the WWRF are outlined in the RWD or in Title 22 criteria:

- | <u>Setback Type</u>                   | <u>Distance (feet)</u> |
|---------------------------------------|------------------------|
| Residential property line             | 150                    |
| Agricultural property line            | 100                    |
| Domestic water supply well            | 100                    |
| Agricultural (irrigation) supply well | 100                    |
12. Objectionable odors originating at this facility shall not be perceivable beyond the limits of the wastewater treatment and recycled water applications areas.
  13. As a means of discerning compliance with Discharge Specification No. 12, the dissolved oxygen content in the upper zone (one foot) of all wastewater ponds shall not be less than 1.0 mg/l.
  14. The discharge to the recycled water areas shall not have a pH less than 6.5 or greater than 8.5.
  15. The wastewater ponds shall be managed to prevent breeding of mosquitoes. In particular,
    - a. An erosion control program shall assure that small coves and irregularities are not created around the perimeter of the water surface.
    - b. Weeds shall be minimized through control of water depth, harvesting, or herbicides.
    - c. Dead algae, vegetation, and debris shall not accumulate on the water surface.
  16. Public contact with wastewater in the recycled water ponds shall be precluded through such means as fencing, signs and other acceptable alternatives.
  17. No waste constituent shall be released or discharged, or placed where it will be released or discharged, in a concentration or mass that causes violation of the Interim Groundwater Limitations.

**C. Recycled Water Specifications**

1. Recycled water shall meet the criteria contained in Title 22, Division 4, CCR (Section 60301, et seq.).
2. Recycled water controllers, valves, etc. shall be affixed with recycled water warning signs, and these and quick couplers and sprinkler heads shall be of a type, or secured in such a manner, that permits operation by authorized personnel only.
3. All recycled water pipes and appurtenances, including but not limited too, valve casing, valve boxes and sprinkler heads, must be color-coded using purple pipe and appurtenances. Purple identification tapes should not be used unless purple materials cannot be obtained.

4. Recycled water treated in accordance with Section 60304(a), Article 4, Division 4, Title 22, CCR may be discharged in the following designated areas: the recycled water storage pond, emergency storage tank, golf course and approved landscape areas, as shown on Attachment F.
5. The turbidity of the influent to the filtration process shall not exceed 10 NTUs.
6. Recycled water may not be discharged 24 hours before precipitation, during periods of precipitation, and for at least 24 hours after cessation of precipitation, or when soils are saturated.
7. The following setbacks for the recycled water application areas were proposed in the RWD and/or have been approved by DHS.

|                                       |          |
|---------------------------------------|----------|
| Domestic water supply well            | 100 feet |
| Agricultural (irrigation) supply well | 50 feet  |
8. Direct or windblown recycled water spray, mist or runoff shall be confined to the designated recycled water application areas and prevented from entering dwellings, designated outdoor eating areas, or food handling facilities.
9. Drinking water fountains shall be protected against contact with recycled water spray, mist, or runoff.
10. Public contact with recycled water shall be precluded through such means as fences, signs, and/or other acceptable alternatives.
11. To alert the public of the use of recycled water, signs with the wording **“RECYCLED WATER – DO NOT DRINK”** and of sufficient size shall be placed at areas of access and around the perimeter of all the water recycling application areas.
12. The water recycling application areas shall be managed to prevent breeding of mosquitoes. More specifically:
  - a. All applied recycled water must infiltrate completely within 24 hours.
  - b. Ditches not serving as wildlife habitat should be maintained free of emergent, marginal, and floating vegetation.
  - c. Low-pressure and un-pressurized pipelines and ditches, which are accessible to mosquitoes, shall not be used to store recycled water.
13. Supplementing recycled water by connection with domestic drinking water source or irrigation or industrial wells requires an air gap device.

**D. General Solids Disposal Specifications:**

Sludge, as used in this document, means the solid, semisolid, and liquid residues removed during primary, secondary, or advanced wastewater treatment processes. Solid waste refers to grit and screenings generated during preliminary treatment. Residual sludge means sludge that will not be subject to further treatment at the WWRF. Biosolids refers to sludge that has been treated and tested and shown to be capable of being beneficially and legally used pursuant to federal and state regulations as a soil amendment for agriculture, silviculture, horticulture, and land reclamation activities.

1. Sludge and solid waste shall be removed from screens, sumps, ponds and clarifiers as needed to ensure optimal plant operation.
2. Treatment and storage of sludge generated by the WWRF shall be confined to the WWRF property and shall be conducted in a manner that precludes infiltration of waste constituents into soils in a mass or at concentration that will violate Groundwater Limitations of this Order.
3. Any storage of residual sludge, solid waste, and biosolids on property of the WWRF shall be temporary and shall be controlled and contained in a manner that minimizes leachate formation and precludes infiltration of waste constituents into soils in a mass or at concentrations that will violate Groundwater Limitations of this Order.
4. Residual sludge, biosolids, and solid waste shall be disposed of in a manner approved by the Executive Officer and consistent with Title 27. Removal for further treatment, disposal, or reuse at sites (i.e, landfill, WWRF, composting sites, soil amendment sites) operated in accordance with valid waste discharge requirements issued by a Regional Water Quality Control Board will satisfy this specification.
5. Use of biosolids as a soil amendment shall comply with valid waste discharge requirements issued by a Regional Water Quality Control Board. In most cases, this will mean the General Biosolids Order (State Board Water Quality Order No. 2000-10-DWQ, *General Waste Discharge Requirements for the Discharge of Biosolids to Land for Use as a Soil Amendment in Agricultural, Silvicultural, Horticultural, and Land Reclamation Activities*). For a biosolids use project to be covered by the General Biosolids Order, the Discharger must file a complete Notice of Intent and receive a Notice of Applicability for each project.
6. Use and disposal of biosolids should comply with the self-implementing federal regulations of Title 40, Code of Federal Regulations (CFR), Part 503, which are subject to enforcement by the U.S. Environmental Protection Agency (EPA), not the Regional Board. If, during the life of this Order, the State accepts primacy for implementation of 40 CFR 503, the Regional Board may also initiate enforcement where appropriate.

## E. Ground Water Limitations:

Release of waste constituents from any storage, treatment, or disposal component associated with the WWRF shall not cause groundwater under and beyond the facility and discharge area(s), as determined by an approved well monitoring network, to:

1. Contain any of the following constituents in concentration greater than the water quality objective or greater than background groundwater quality, whichever is greater:

| <u>Constituent</u>   | <u>Units</u> | <u>Interim Limitation</u>      |                                |
|----------------------|--------------|--------------------------------|--------------------------------|
|                      |              | <u>Water Quality Objective</u> | <u>Background</u> <sup>1</sup> |
| Boron                | mg/l         | 0.6                            | unknown <sup>2</sup>           |
| Chloride             | mg/l         | 106                            | 138                            |
| Iron                 | mg/l         | 0.3                            | ND                             |
| Manganese            | mg/l         | 0.05                           | 0.64                           |
| Sodium               | mg/l         | 69                             | 155                            |
| Total Nitrogen       | mg/l         | 10                             | unknown <sup>2</sup>           |
| as Nitrite (as N)    | mg/l         | 1                              | unknown <sup>2</sup>           |
| as Nitrate (as N)    | mg/l         | 10                             | 18                             |
| as Ammonia (as N)    | mg/l         | 0.5                            | 0.5                            |
| TDS <sup>3</sup>     | mg/l         | 450                            | 1100                           |
| EC                   | µmhos/cm     | 700                            | 1450                           |
| pH                   | pH Units     | 6.5 – 8.5                      | 7.6                            |
| Total Coliform       | MPN/100 mL   | ND                             | ND                             |
| Total Trihalomethane | µg/L         | 100                            | unknown <sup>2</sup>           |

<sup>1</sup> Four groundwater samples were collected in August 2001 from the test boreholes HB-2, HB-7, HB-3 and B-27 as shown on Attachment F.

<sup>2</sup> Analysis for this compound was not performed during the August 2001 investigation for shallow groundwater quality.

<sup>3</sup> A cumulative impact limit that accounts for several dissolved constituents in addition to those listed here separately [e.g., alkalinity (carbonate and bicarbonate), calcium, hardness, phosphate, potassium, etc.].

2. Contain any constituent not identified in Interim Groundwater Limitation E.1 in concentrations greater than background quality (whether chemical, physical, biological, bacteriological, radiological, or some other property or characteristic).
3. Impart taste, odor, or color that creates nuisance or impairs any beneficial use.

**F. Surface Water Limitations**

The discharge shall not cause the surface watercourse downstream of the recycled water storage pond, or recycled water areas to contain waste constituents in concentrations statistically greater than background (i.e., upstream) surface water quality.

**G. Provisions:**

1. By **15 July 2002**, the Discharger shall submit a NOI or a Notice of Non-Applicability for coverage under the NPDES General Permit No. CAS000001, *Discharges of Storm Water Associated with Industrial Activities Excluding Construction Activities*.
2. All of the reports as required by this Order shall be submitted pursuant to Section 13267 of the California Water Code.
3. At least **60 days** prior to discharge of wastewater to the WWRF, the Discharger shall submit the following and implement the required measures as appropriate:
  - a. A report prepared by a California Registered Engineer that describes potential scenarios of wastewater treatment system failures, and the costs to remedy such failures. **Within 30 days** of Regional Board staff's written acceptance of this report, the Discharger shall provide for this amount or a financial assurance of this amount to be placed into the reserve fund for the WWTF. A letter shall be submitted to the Executive Officer from the responsible financial agent or agency to verify that that this has been done **within 30 days** of the deposit of funds or financial assurance mechanism. The financial assurance must remain available to the Discharger for repairs until such time that the Executive Officer indicates the financial assurances is no longer needed. The Discharger shall notify the Regional Board annually **by 1 January** that the financial assurance mechanism remains in place.
  - b. A signed contract for the operation, maintenance and monitoring of the wastewater treatment plant. The contractor performing these functions must be a Registered Engineer, Registered Environmental Health Specialist or Certified Treatment Plant Operator. Any modifications to the contract or transfer of the contract to another person or agency shall be subject to approval of the Executive Officer.
  - c. A report by a California Registered Civil Engineer that estimates the annual amount of funds necessary to accrue an amount of capital sufficient to replace the treatment and disposal system at the end of its useful life. The Discharger shall then establish an account for these funds and shall file an annual report with the Regional Board **by 1 January** of each year verifying the account contains adequate capital in accordance with the engineer's report.

4. Within **30 days** of discharging wastewater to the WWRF, the Discharger shall submit a report showing that it has installed and accurately calibrated an influent wastewater flow monitoring device, such that peak dry weather and peak wet weather flows can be recorded daily. Demonstration of accurate calibration for the continuous monitoring of turbidity and chlorine residual shall also be provided.
5. Prior to the discharge of any wastewater to the WWRF the Discharger shall complete and maintain an Operations and Maintenance (O & M) Manual, which shall be kept current of the existing conditions and procedures at the WWRF. Operating personnel shall keep a copy of the O & M Manual at the wastewater treatment plant site for reference. Operating personnel shall be familiar with its contents and location, and it shall be made available for review during Regional Board inspection. At a minimum, items to be kept with the O & M Manual are as follows:
  - a. Site plan with description of the treatment components including operating procedures for each component, piping diagrams and process schematics, discussion of alarms and alarm response, discussion of maintenance and inspection procedures,
  - b. Steps to take when there is a problem with the system, operation and maintenance records to include equipment maintenance schedules and operator logs,
  - c. Monitoring and Reporting Program self-monitoring reports, system upsets, sludge disposal, and personnel training,
  - d. An emergency response plan to include reporting procedures, agency contact phone numbers, a spill response plan for hazardous materials, raw wastewater, partially treated wastewater, spill cleanup, and cleanup waste products disposal plan; and
  - e. A copy of this Order.

Engineer stamped as-built drawings that should also be submitted to the Regional Board.

An engineer's report documenting that the treatment plant, disposal area and all items auxiliary to the plant have been constructed in accordance with the plans and specifications.

The report shall include:

- i. All leak testing information, including all piping and tanks,
  - ii. An inspection log verifying an inspector was present for all critical phases of construction,
  - iii. Logs of pump testing, filter testing, blower testing, and testing of any other equipment,
  - iv. Testing of disposal area for even distribution of wastewater,
  - v. Certification that all construction complies with County Codes.
6. The following items shall be submitted to DHS for their review and approval:
  - Prior to the construction of the WWRF, the Discharger shall submit detailed plans and specifications for the tertiary filtration facility, including the backwash and coagulation systems.

- A copy of the operations plan and contingency plan shall be submitted once the WWRF is in operation.
  - A report demonstrating that the minimum disinfection contact time value is greater than 450 milligram-minutes per liter with a modal contact time of at least 90 minutes, based on peak dry weather design flow in accordance with CCR Title 22, Section 60301.230. The demonstration can consist of calculations based on the dimensions of the chlorine contact basin(s), or a tracer study on the completed facility.
7. By **15 July 2002**, the Discharger shall submit a Groundwater Monitoring Well Installation Workplan. The Workplan shall be prepared by a Registered Civil Engineer or Registered Geologist and shall contain the information listed in the first section of Attachment G. The Workplan shall propose a monitoring network consisting of one or more background monitoring wells, as well as compliance monitoring wells immediately downgradient of treatment, storage, and disposal units that do or may release waste constituents to groundwater. Every monitoring well shall be constructed to yield representative samples from the uppermost layer of the uppermost aquifer and shall comply with applicable DWR Well Standards.
  8. By **1 November 2002**, the Discharger shall submit a Monitoring Well Installation Report prepared by a Registered Civil Engineer or a Registered Geologist. The report shall describe the installation of groundwater monitoring wells and shall contain the items identified in Attachment G. The Report shall provide the initial results of analysis performed to confirm or to augment background groundwater quality.
  9. By **1 November 2002**, the Discharger shall submit a Sludge Disposal Plan. The plan shall provide a detailed program and time schedule for permanent disposal of all solid wastes generated at the wastewater treatment plant and tertiary treatment plant. The Sludge Disposal Plan shall include the items listed on Attachment H.
  10. By **1 November 2002**, the Discharger shall submit a Recycled Water Contingency Plan. The plan shall include notification of the Regional Board and health agencies and outline actions to be taken when tertiary effluent quality fails to meet required standards for recycled water use specified in this Order.
  11. By **1 January 2003**, the Discharger shall submit a Golf Course Irrigation Management Plan (GCIMP). The GCIMP shall outline what measures the Discharger will incorporate to manage the application of recycled water such that recycled water does not co-mingle with storm water and be discharged to surface water or surface water drainage courses. The golf course may use course design, engineering controls and/or operational controls to prevent the discharge of recycled water or co-mingled recycled water and stormwater to surface water or surface water drainage courses.



12. By **1 February 2003**, the Discharger shall submit a *Sanitary Sewer Collection Operation, Maintenance, Overflow Prevention, and Response Plan* (SSO Plan) that describes the actions designed to prevent, or minimize the potential for sanitary sewer overflows. The Discharger shall maintain the SSO Plan in an up-to-date condition and shall amend the SSO Plan whenever there is a change (e.g., in the design, construction, operation, or maintenance of the sanitary sewer system or sewer facilities) that materially affects the potential for sanitary sewer overflows, or whenever there is a sanitary sewer overflow. The Discharger shall ensure that the up-to-date SSO Plan is readily available to sewer system personnel at all times and that sewer system personnel are familiar with it.

At a minimum, the Collection System Operation and Maintenance portion of the plan shall contain or describe the following:

- Detailed maps of the sanitary sewer system, identifying sewer mains, manholes, and lift stations;
- A detailed listing of elements to be inspected, a description of inspection procedures and inspection frequency, and sample inspection forms;
- Routine inspection and testing of all pipelines, lift stations, valves, and other key system components. The inspection/testing program shall be designed to reveal problems that might lead to accidental spills and ensure that preventive maintenance is completed;
- Repair or replacement of old, worn out, or defective equipment;
- Minimize the need for manual operation of critical systems and provide spill alarms or other “fail safe” mechanisms;
- Properly manage, operate and maintain, at all times, all parts of the collection system that the Discharger owns or over which the Discharger has operational control;
- Provide adequate capacity to convey base flows and peak flows for all parts of the collection system the Discharger owns or over which the Discharger has operational control; and
- Take all feasible steps to stop and mitigate the impact of sanitary sewer overflows in portions of the collection system the Discharger owns or over which the Discharger has operational control; and

At a minimum, the Overflow Prevention and Response Plan shall contain or describe the following:

- Identification of areas of the collection system that historically have overflowed and an evaluation of the cause of the overflow;
- Maintenance activities that can be implemented to address the cause of the overflow and means to prevent future overflows. Maintenance activities may include pretreatment of wastewater from industrial dischargers who discharge high concentrations of oil and grease in their wastewater;

- Procedures for responding to sanitary sewer overflows designed to minimize the volume of sewer overflow that enters surface waters, and minimize the adverse effects of sewer overflows on water quality and beneficial uses;
  - Describe steps to be taken when an overflow or spill occurs, and procedures that will be implemented to ensure that all overflows and spills are properly identified, responded to and reported; and
  - A public notification plan, in which any posting of areas contaminated with sewage is performed at the direction of the Yolo County Health Department. All parties with a reasonable potential for exposure to an overflow event shall be notified.
13. By **1 May 2004** the Discharger shall submit a written technical report, prepared by a California Registered Civil Engineer or Registered Geologist, that characterizes the groundwater quality of each monitoring well. For each monitoring parameter/constituent outlined in the MRP, the report shall compare measured concentrations for compliance monitoring wells with: 1) the calculated background concentration, and 2) the interim numeric limitations set forth in Groundwater Limitation E.1.a. Where background concentrations are statistically greater than the interim limitations specified in Groundwater Limitation E.1.a, the report shall recommend final groundwater limitations for waste constituents listed therein. Subsequent use of a concentration as a final groundwater limitation will be subject to the discretion of the Executive Officer. Where background concentrations are statistically less than the listed interim concentrations and the concentrations in the downgradient wells are statistically greater than background concentrations, the Discharger shall begin the preparation of a BPTC Evaluation Report to be approved by the Executive Officer.

### **BPTC EVALUATION REPORT**

14. Provisions G.14.a through G.14.d outline the schedule and compliance dates to be adhered to if the Discharger is required to submit a BPTC Evaluation Report per Provision G.13:
- a. By **1 May 2005**, the Discharger shall submit for Executive Officer approval a written work plan in the form of a technical report that sets forth a schedule for systematic and comprehensive technical evaluation of each component of the WWRF's waste treatment and control to determine for each waste constituent BPTC as used in Resolution 68-16. The technical report shall contain a preliminary evaluation of each component and propose a time schedule for completing the comprehensive technical evaluation. The technical report shall be prepared and certified by a California Registered Civil Engineer. The schedule to complete all comprehensive technical evaluations shall be as short as practicable and shall not exceed one year.
  - b. By the schedule approved by the Executive Officer the Discharger shall submit by **1 May 2007**, the written comprehensive technical evaluation shall be submitted with the Discharger's written recommendations for the WWRF modifications (i.e., component

- upgrade and retrofit). The report shall include specific methods the Discharger proposes as a means to measure processes and assure continuous optimal performance of BPTC measures. Comprehensive technical evaluations shall be prepared and certified by a California Registered Civil Engineer. The source of funding and proposed schedule for modifications shall be identified. The schedule shall be as short as practicable but in no case shall exceed five years past the Executive Officer's approval of the comprehensive technical evaluation unless the schedule is reviewed and specifically approved by the Regional Board. The component evaluation, recommended improvements and schedule are subject to the Executive Officer's review and approval.
- c. By **1 August 2007**, the Discharger shall submit a technical report that proposes specific numeric groundwater limitations that reflect full implementation of BPTC and describe how these were determined considering actual data from compliance monitoring wells, impact reductions through full implementation of BPTC, reasonable growth, etc. The Discharger should submit results of a validated groundwater model to support its proposal.
  - d. By **1 August 2008**, the District shall submit a technical report on the overall status of compliance with implementation of BPTC and compliance with all groundwater limitations.
- 15. The Discharger shall comply with Monitoring and Reporting Program No. R5-2002-0077.
  - 16. The Discharger shall comply with the "Standard Provisions and Reporting Requirements for Waste Discharge Requirements", dated 1 March 1991, which are attached hereto and by reference a part of this Order. This attachment and its individual paragraphs are commonly referenced as "Standard Provision(s)."
  - 17. At least 90 days prior to termination or expiration of any lease, contract, or agreement involving the disposal or reclamation areas, used to justify the capacity authorized herein and assure compliance with this Order, the Discharger shall notify the Regional Board in writing of the situation and of what measures have been taken or are being taken to assure full compliance with this Order.
  - 18. The Discharger shall submit to the Regional Board on or before each compliance report due date the specified document, or if appropriate, a written report detailing compliance or noncompliance with the specific schedule date and task. If noncompliance is reported, then the Discharger shall state the reasons for noncompliance and shall provide a schedule to come into compliance.
  - 19. The Discharger shall use the best practicable cost-effective control technique(s) currently available to comply with discharge limits specified in this Order.

20. The Discharger shall implement best practicable treatment and control, including proper operation and maintenance to comply with this Order.
21. The Discharger shall report promptly to the Regional Board any material change or proposed change in the character, location, or volume of the discharge.
22. In the event of any change in control or ownership of land or waste discharge facilities presently owned or controlled by the Discharger, then the Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be forwarded to this office.
23. The Discharger shall comply with all conditions of this Order, including timely submittal of technical and monitoring reports as directed by the Executive Officer. Violations may result in enforcement action, including Regional Board or court orders requiring corrective action or imposing civil monetary liability, or in revision or rescission of this Order.
24. A copy of this Order shall be kept at the discharge facility for reference by operating personnel. Key operating personnel shall be familiar with its contents.
25. The Regional Board will review this Order periodically and will revise requirements when necessary.

I, GARY M. CARLTON, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 26 April 2002.

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GARY M. CARLTON, Executive Officer